

An MLP-Based Safeguard for Hermite-Basis VDF Reconstruction

Removing Gibbs-type artifacts from truncated Hermite decompositions of strongly non-Maxwellian velocity distribution functions in a magnetic reconnection simulation

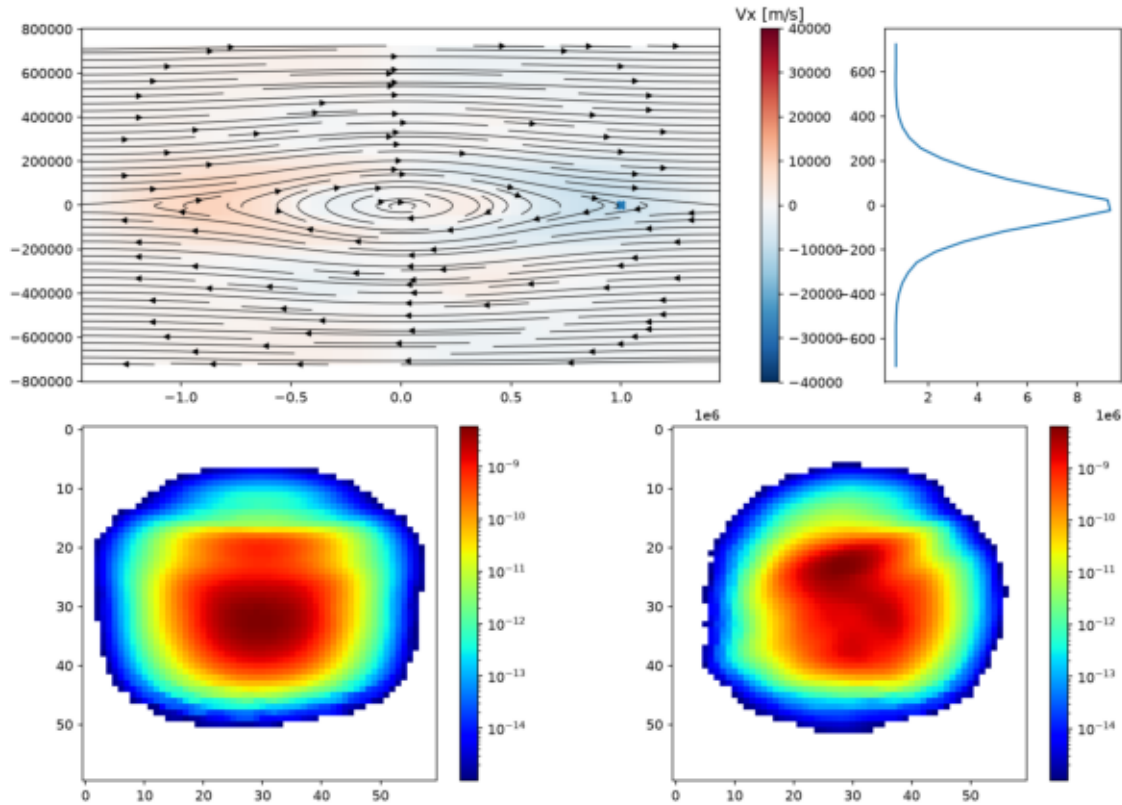
Vlasiator 2D GEM-challenge reconnection run — reconnection_2d_beta025

Outline

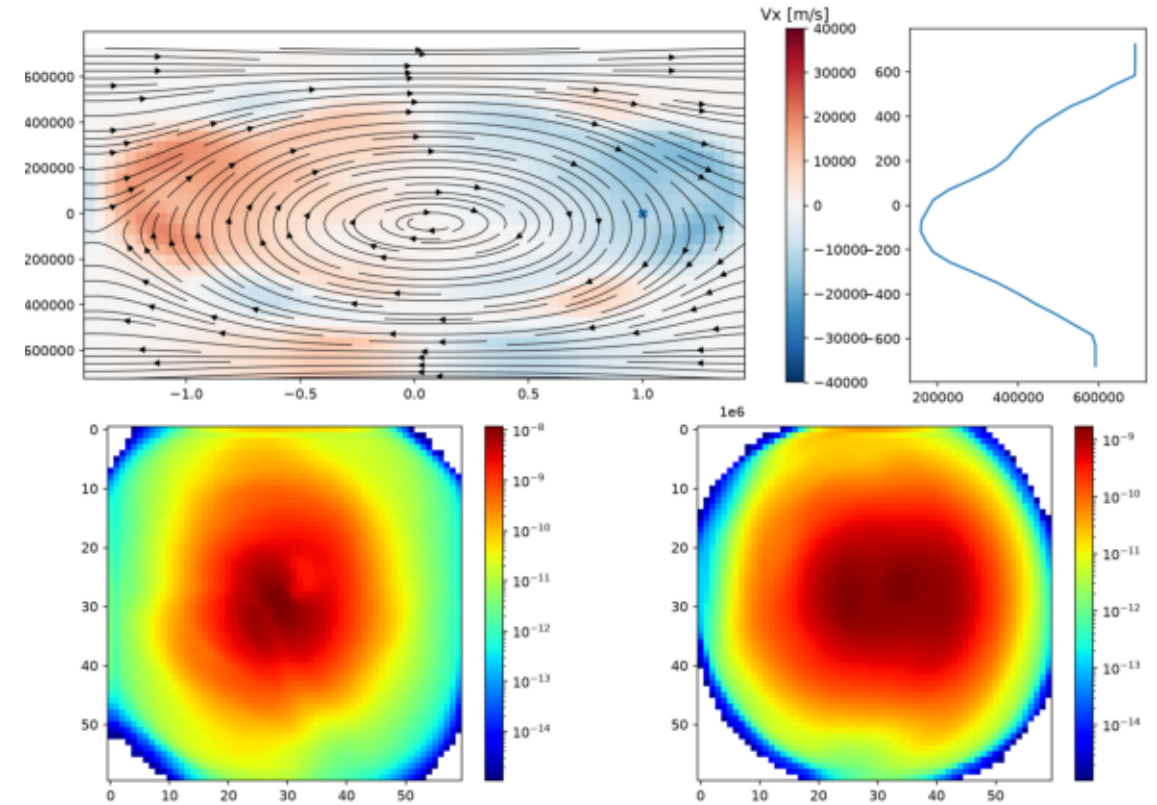
- 1. The simulation: 2D magnetic reconnection (GEM challenge)
- 2. Velocity distribution functions in the current layer
- 3. Hermite-basis reconstruction and its Gibbs-artifact problem
- 4. ML corrector design: $f_{\text{final}} = f_{\text{rec}} * \exp(\Delta_{\text{pred}})$
- 5. Feature evolution: low-S coefficients $\rightarrow$ 1-D spectra $\rightarrow$ local patch
- 6. Within-snapshot results
- 7. Cross-timestep generalization: failure and fix
- 8. Conclusions and open weaknesses

1. The Simulation: 2D Magnetic Reconnection

GEM-challenge current sheet, $\beta=0.25$ — V_x colormap with B field lines, ion density profile, and a sample VDF at (1000 km, 0, 0)



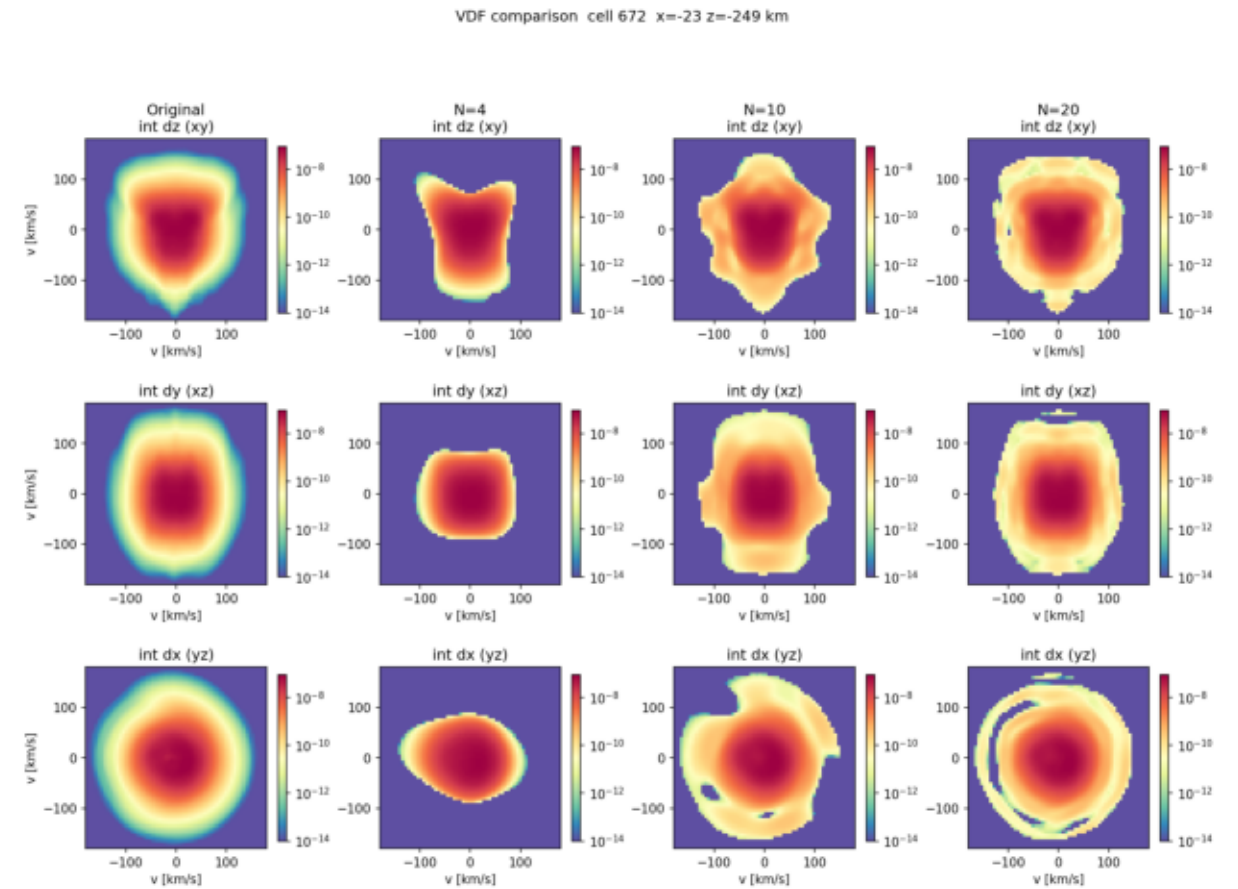
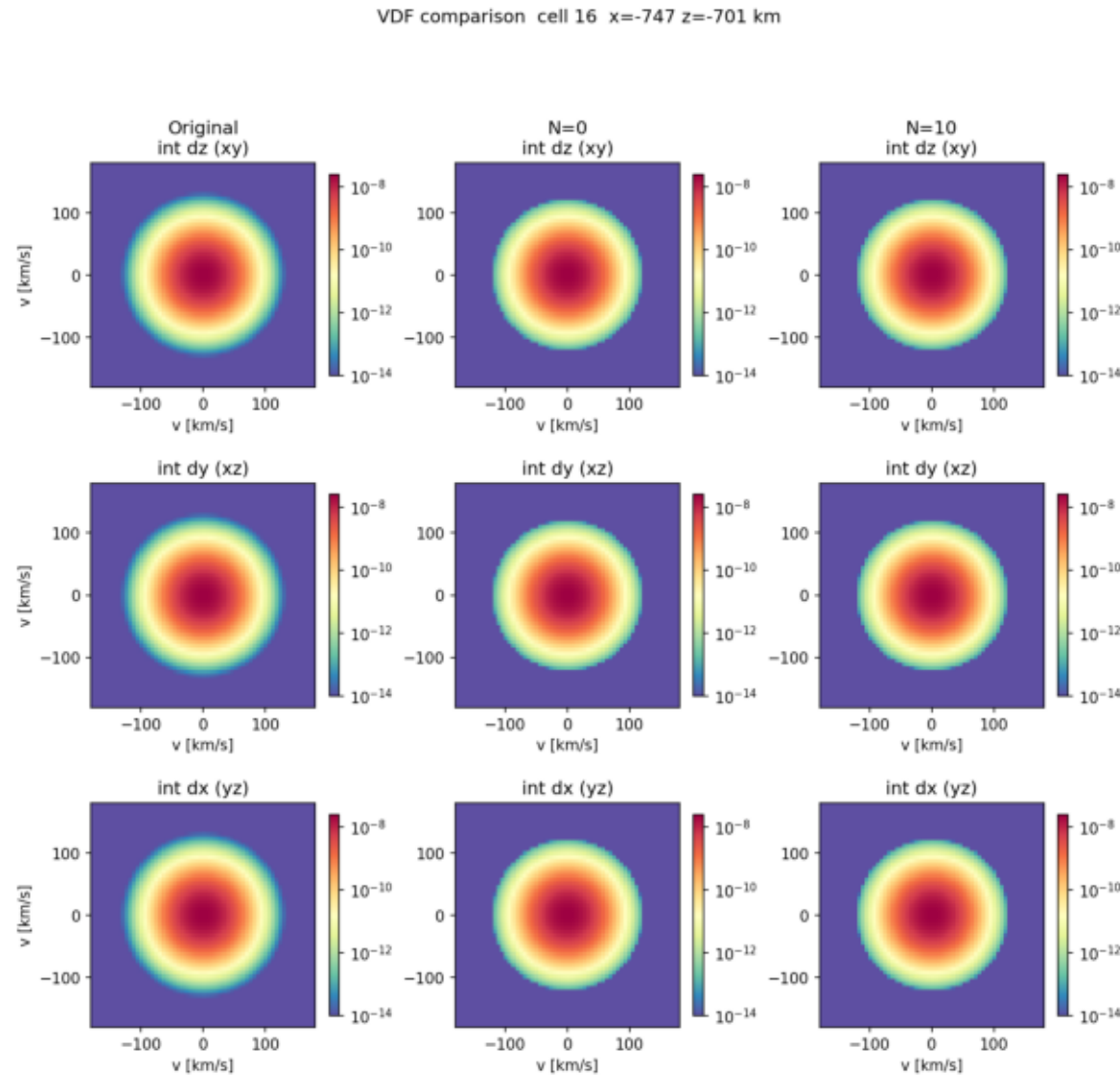
Early time (bulk.0000022) — thin, still-forming current sheet



Late time (bulk.0000111) — fully developed reconnection outflows

2. VDFs in the Current Layer

Two representative cells from bulk.0000024: a lobe cell (near-Maxwellian) vs. the current-sheet center (strongly non-Maxwellian)



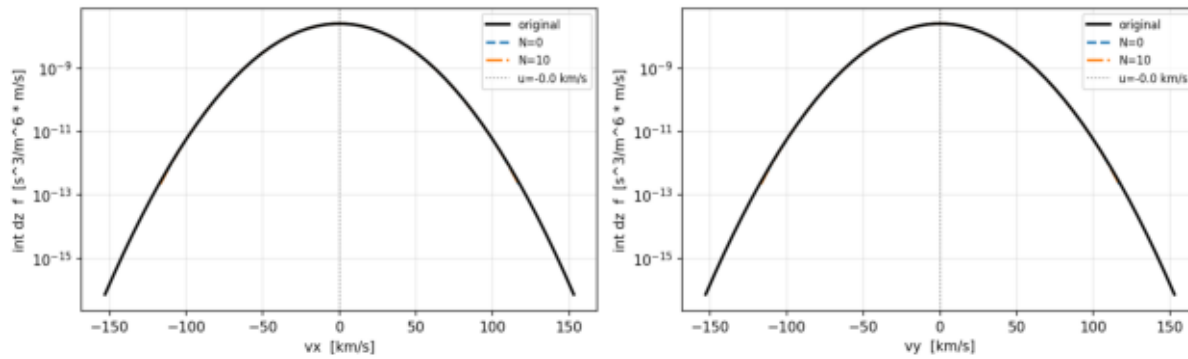
cid=672 — current-sheet center

cid=16 — lobe, far from the current sheet

2. VDFs in the Current Layer (cont.)

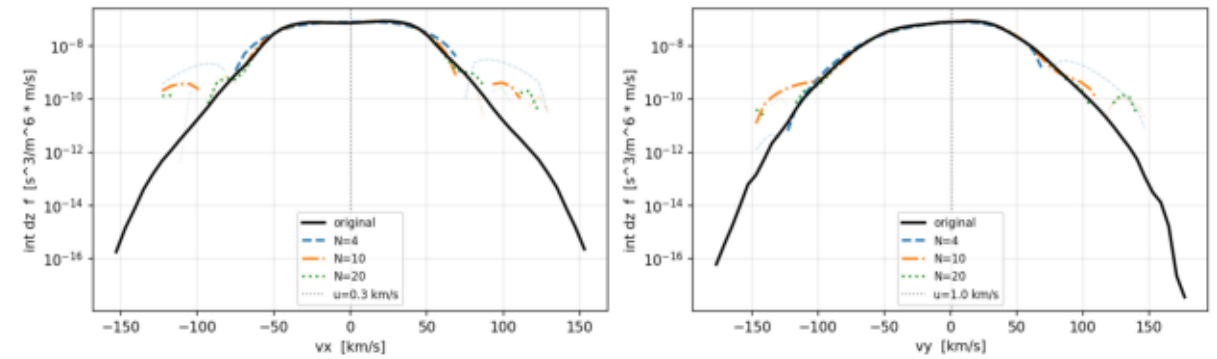
1-D cuts through the integrated VDF, original vs. Hermite reconstruction at increasing order — note the ringing that appears for the non-Maxwellian

1-D cuts of int dz VDF cell 16 $x=-747$ $z=-701$ km



cid=16 (lobe) — converges cleanly

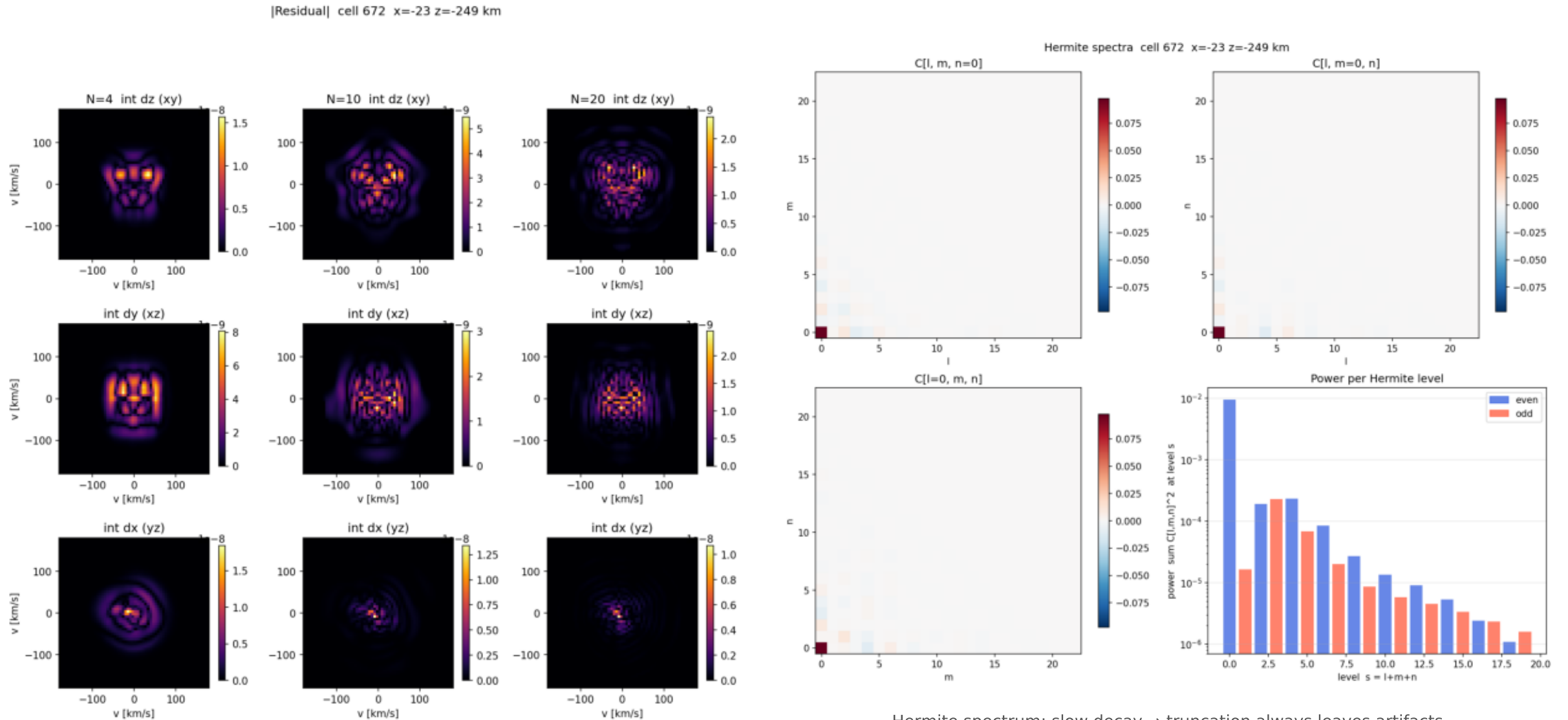
1-D cuts of int dz VDF cell 672 $x=-23$ $z=-249$ km



cid=672 (current sheet) — Gibbs oscillations at low order

3. The Gibbs-Artifact Problem

Residuals and Hermite-space power spectra for the current-sheet cell (cid=672)

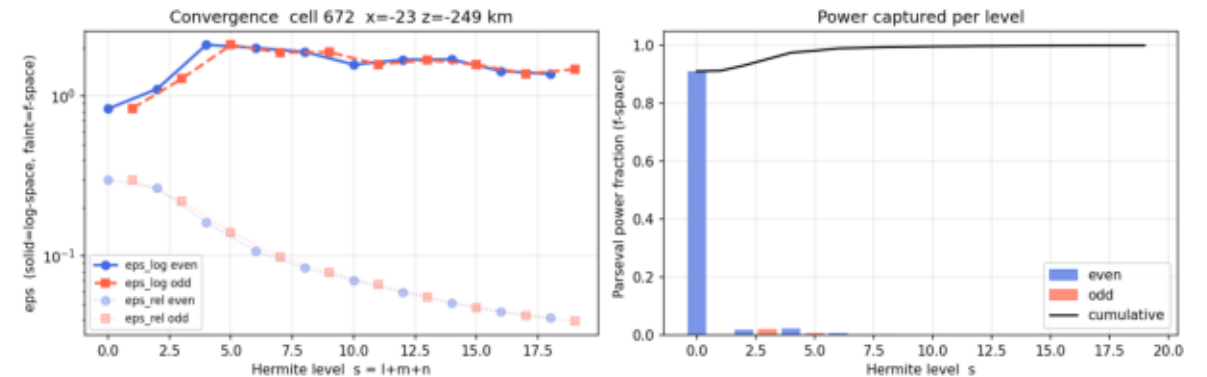


[reconstruction - truth]: ringing concentrated at steep gradients

3. The Gibbs-Artifact Problem (cont.)

Even at high order, 20–38% of in-support voxels go negative in log-space; more orders alone do not remove the ringing

- Truncated tetrahedral Hermite decomposition (up to order 22)
acts as a $\sim 60^3$ velocity-space $\rightarrow \sim 2300$ -coefficient compression
- Sign-flip (Gibbs) ringing appears near steep gradients /
turning points
- `sparse_mask` only zeroes the true-empty region — it does not
fix internal sign flips inside the support
- f-space L2 error (`eps_rel`) is peak-dominated and blind to
tail/log accuracy — `eps_log` (RMS of log-space residual) is
the honest metric
- Conclusion: need a learned, structural correction on top of
the Hermite compression, not just a bigger truncation order



Parseval convergence: `eps_rel` keeps improving while `eps_log` does not

4. ML Corrector Design

- Variant B: an MLP predicts a per-voxel LOG-space delta correction
 - $f_{\text{final}}(v) = f_{\text{rec_full}}(v) \cdot \exp(\Delta_{\text{pred}}(v))$ inside `sparse_mask`, else 0 (structural)
- Multiplicative form guarantees positivity regardless of network output — no clipping, no post-hoc fix-up needed
- Still "compression": the full tetrahedral decomposition (order 22) IS the compression step; the corrector is a small SHARED model that removes artifacts on top of it, conditioned on cheap already-stored low-order coefficients
- Per-cell adaptive regularization on Δ_{pred} :
 - $\lambda_{\text{cell}} = \lambda_{\text{max}} / (1 + (\text{mean_sq_cell} / \text{ref_var})^{\text{power}})$
 - suppresses correction on already-good cells without limiting it on genuinely hard ones

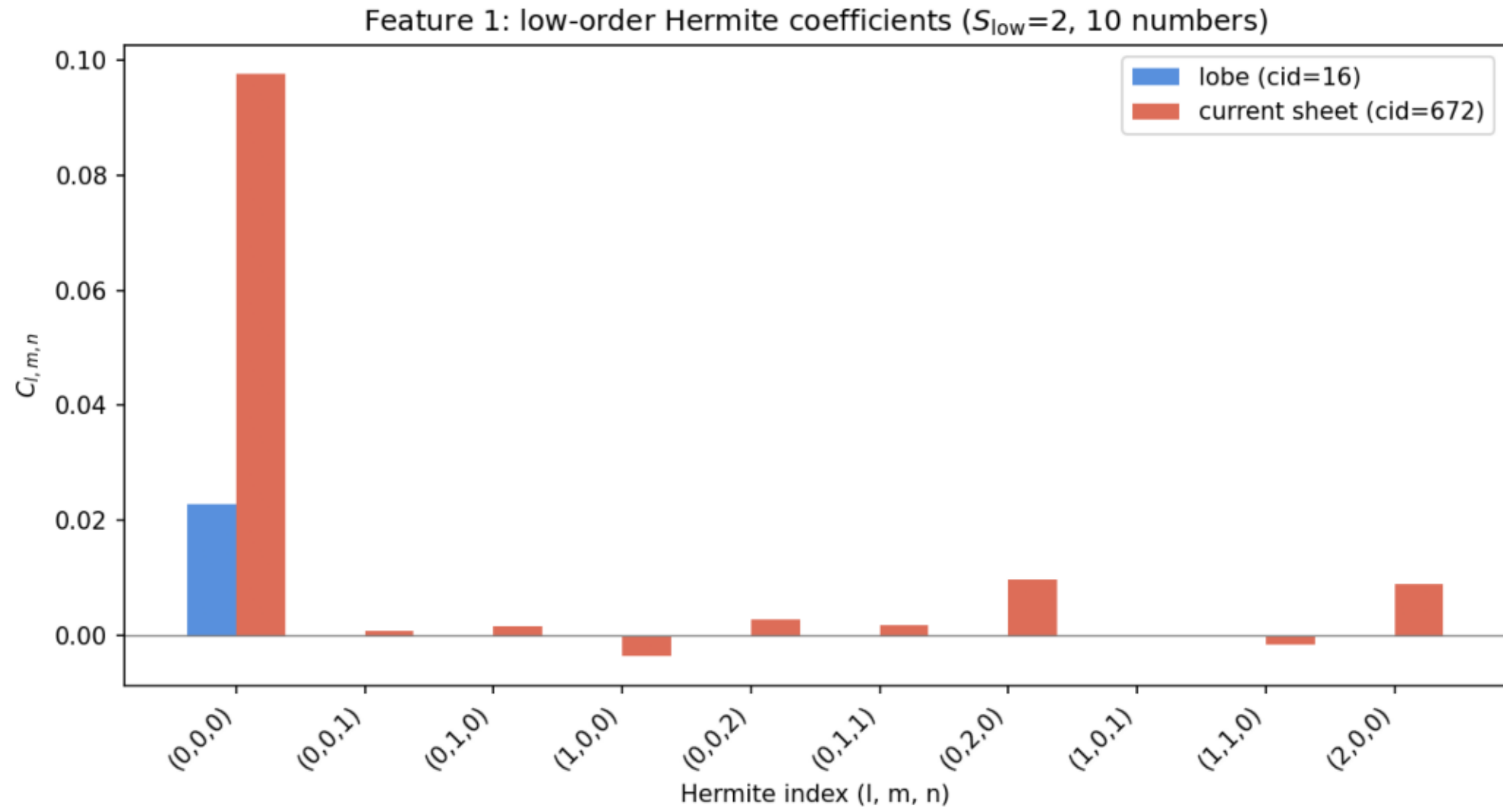
5. Feature Evolution

What information does the MLP need to tell "this looks wrong" from "this is genuinely near-empty"?

- Step 1 — low-S coefficients (C_{low} , $S_{\text{low}} \leq 2$, 10 numbers): global but very coarse; cheap, already computed as part of the compressed representation
- Step 2 — 1-D Hermite-space power spectra ($3 \times (\text{order} + 1) = 69$ numbers): cheap (pure reduction over already-computed coefficients); strong global "how non-Maxwellian is this cell" signal
- Step 3 — local $3 \times 3 \times 3$ patch of $\log(f_{\text{rec_full}})$ around the query voxel (27 numbers): gives spatial context to localize and shape the correction, but alone cannot disambiguate a genuinely empty tail from a wrongly-zeroed real density (both look like a flat patch at the sparsity floor)
- Best result: patch + 1-D spectra COMBINED — each covers the other's blind spot

5a. Feature 1 — Low-Order Coefficients

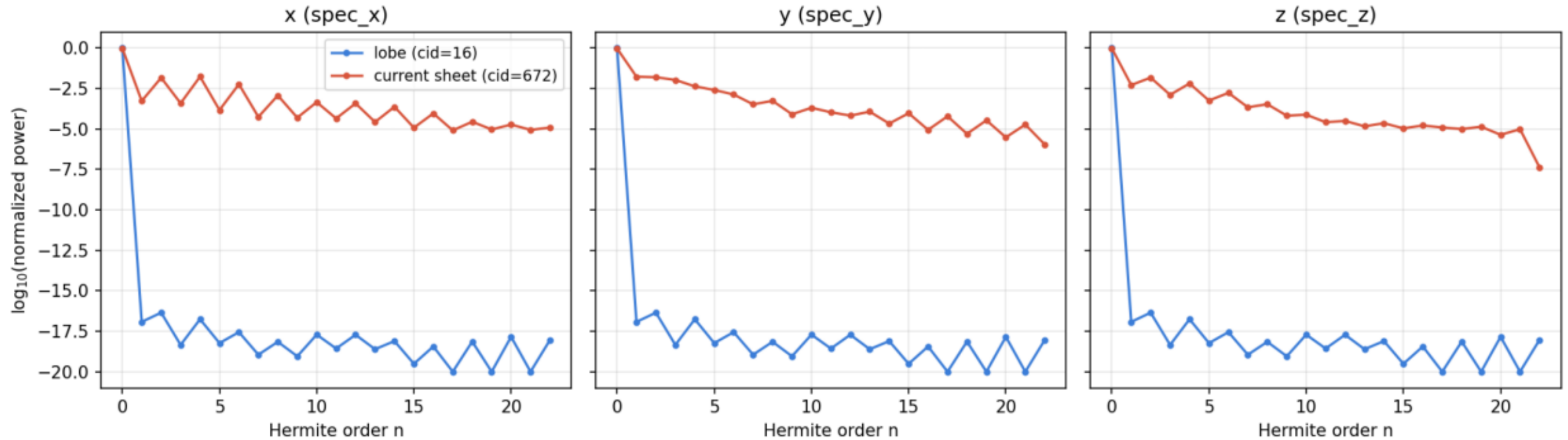
C_low alone is a coarse, global summary — not enough to arbitrate ambiguous per-voxel patch patterns



5b. Feature 2 — 1-D Hermite-Space Power Spectra

$\text{sum}_{l,m} C[l,m,n]^2$ per axis: cheap and gives a strong, clean signal separating Maxwellian-like from strongly non-Maxwellian cells

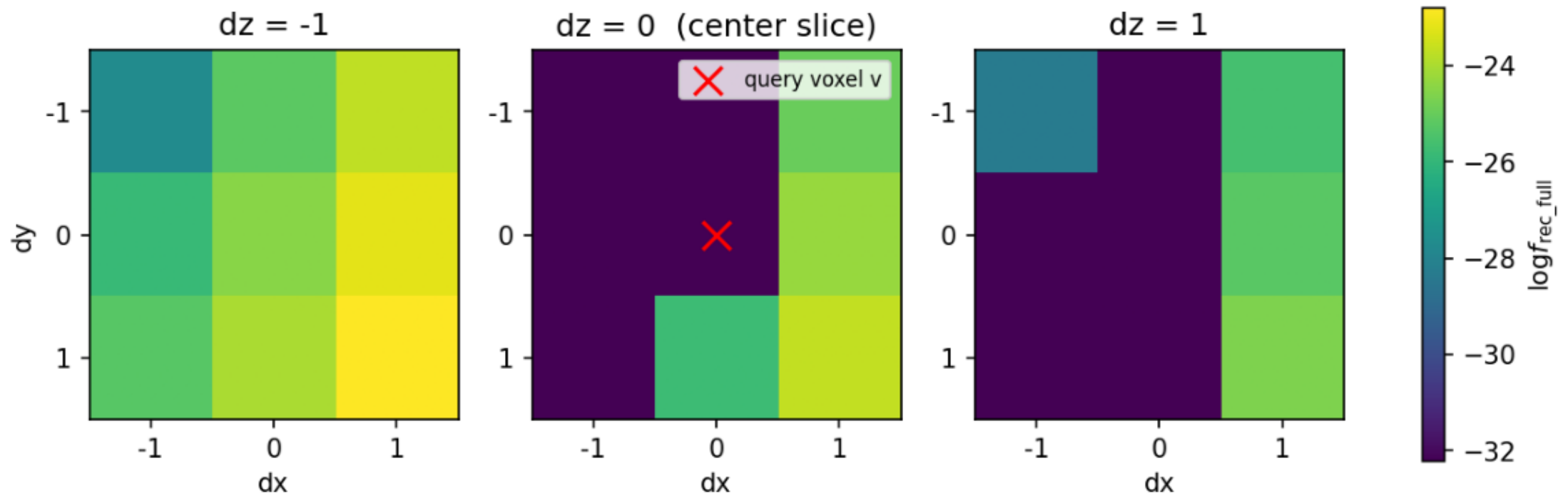
Feature 2: 1-D Hermite-space power spectra ($3 \times (\text{order} + 1) = 69$ numbers)



5c. Feature 3 — Local 3×3×3 Patch

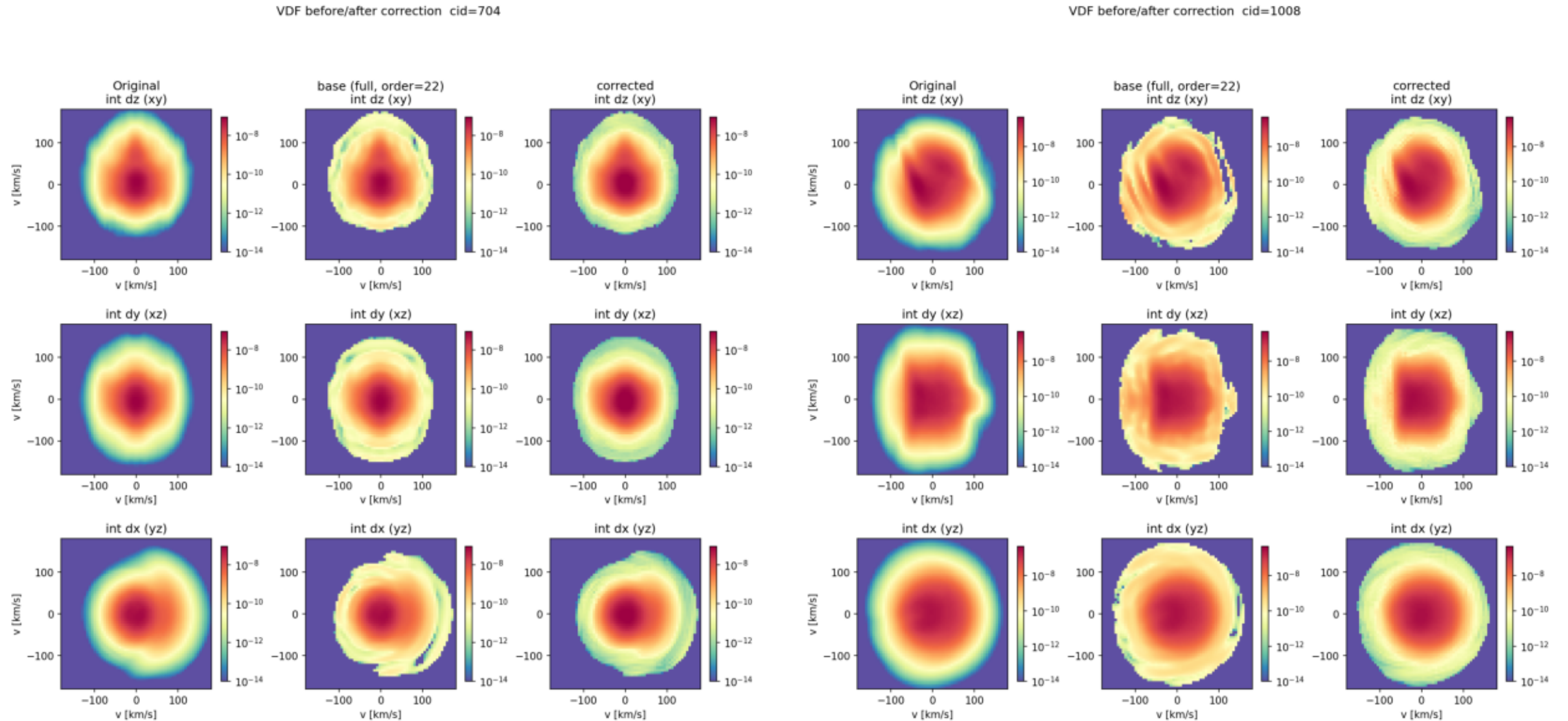
Concatenated into the MLP input (not a real convolution) — gives spatial context to localize Gibbs sign-flip artifacts

Feature 3: local 3x3x3 patch of $\log(f_{\text{rec_full}})$ around a Gibbs-artifact voxel (27 numbers, cid=672)



6. Within-Snapshot Results

Held-out cells of the training bulk file (bulk.0000024) — raw reconstruction (left) vs. corrected (right), 2-D projections

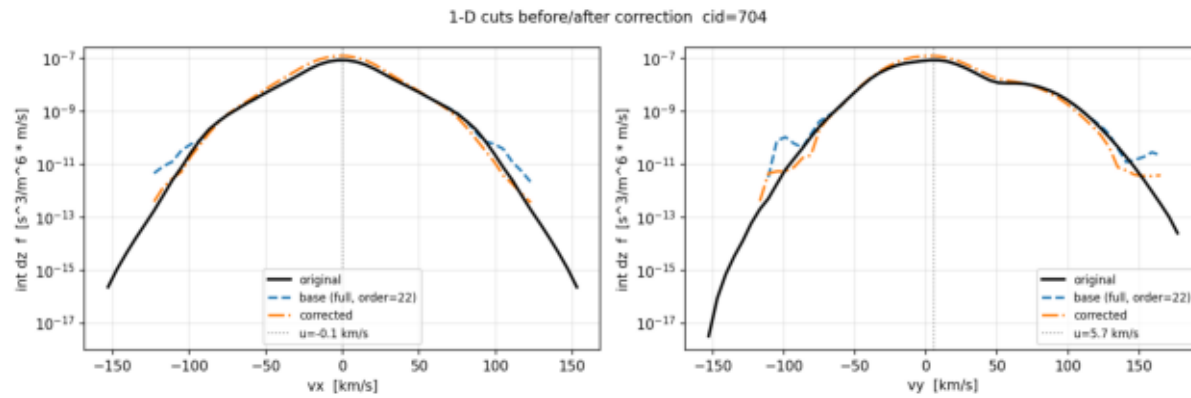


cid=704 — hard, non-Maxwellian cell

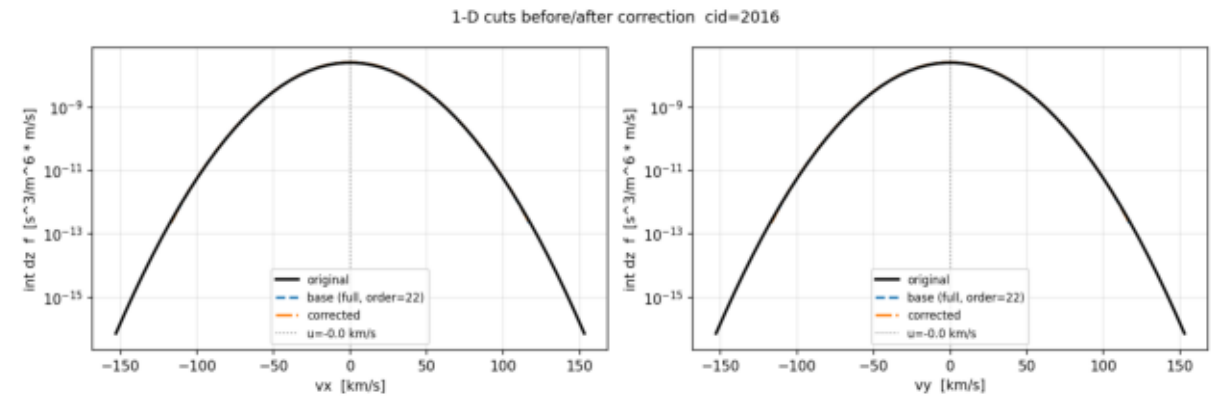
cid=1008

6. Within-Snapshot Results (cont.)

1-D cuts of the integrated VDF — the correction visibly removes ringing while tracking the true profile



cid=704



cid=2016 — near-Maxwellian lobe cell (correction ≈ 0 , as it should)

- Mean improvement on held-out cells (patch + 1-D spectra, current checkpoint): 50.8% reduction in `eps_log`

7. Cross-Timestep Generalization: the Failure

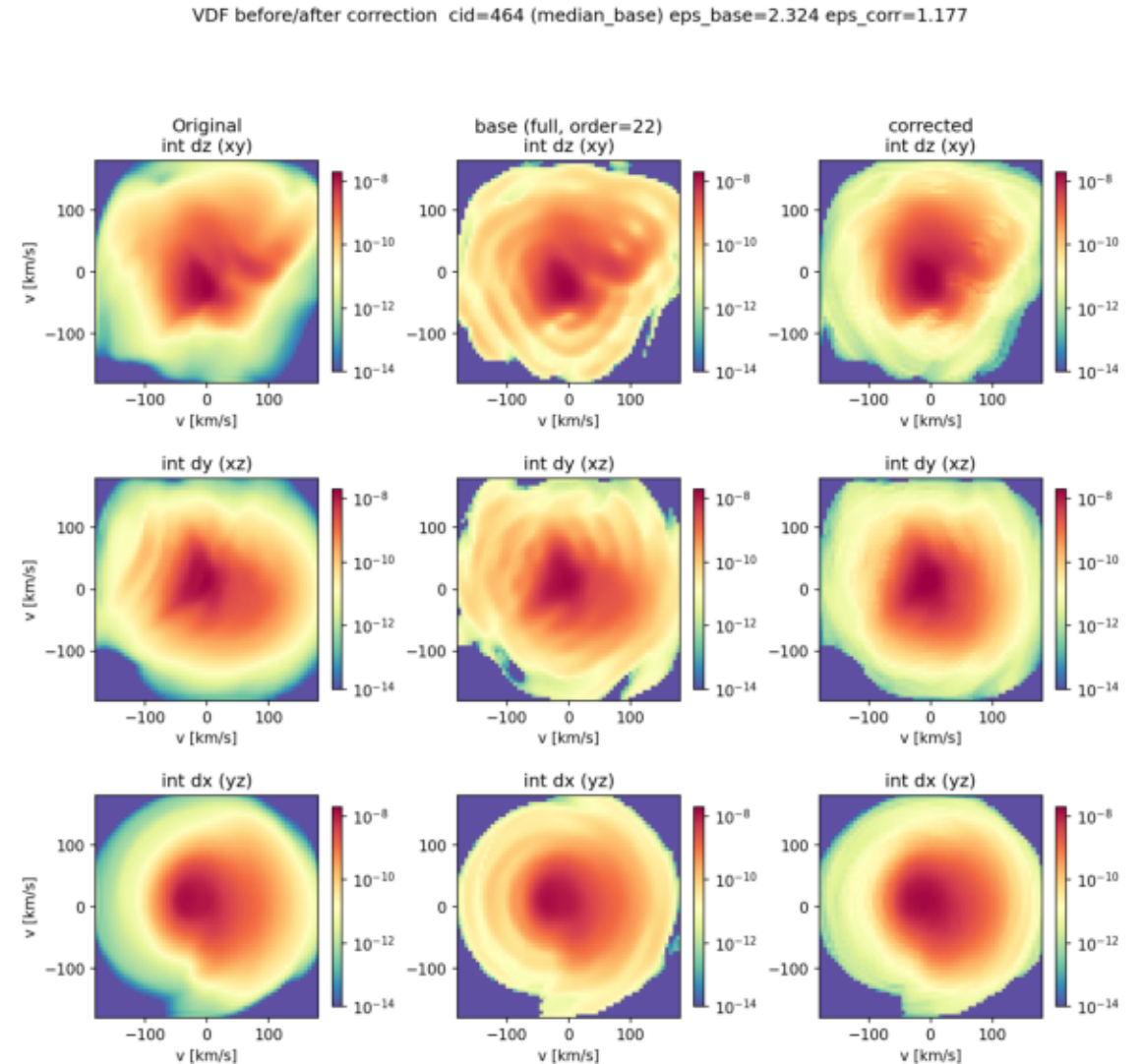
A corrector trained on ONE snapshot (bulk.0000024) evaluated on an unseen timestep (bulk.0000111)

- Within-snapshot (held-out cells, same file): 62.2% mean improvement, near-perfect lobe-cell handling
($\text{eps_log_corr} = 0.0107$)
- Cross-timestep (entirely different, unseen snapshot): mean improvement = -28.0% — the corrector makes things WORSE on average
- Diagnosis: the patch-driven correction overfit to the specific non-Maxwellian signature of the single training snapshot; the axis-spectra "triviality recognition" generalized fine on its own
- Root cause: single-snapshot training data cannot separate a genuine physical pattern from an artifact of that one moment in time

7. Cross-Timestep Generalization: the Fix

Retrained on 3 spread-out timesteps (bulk.0000024/50/80), validated on a 4th, never-seen timestep (bulk.0000111)

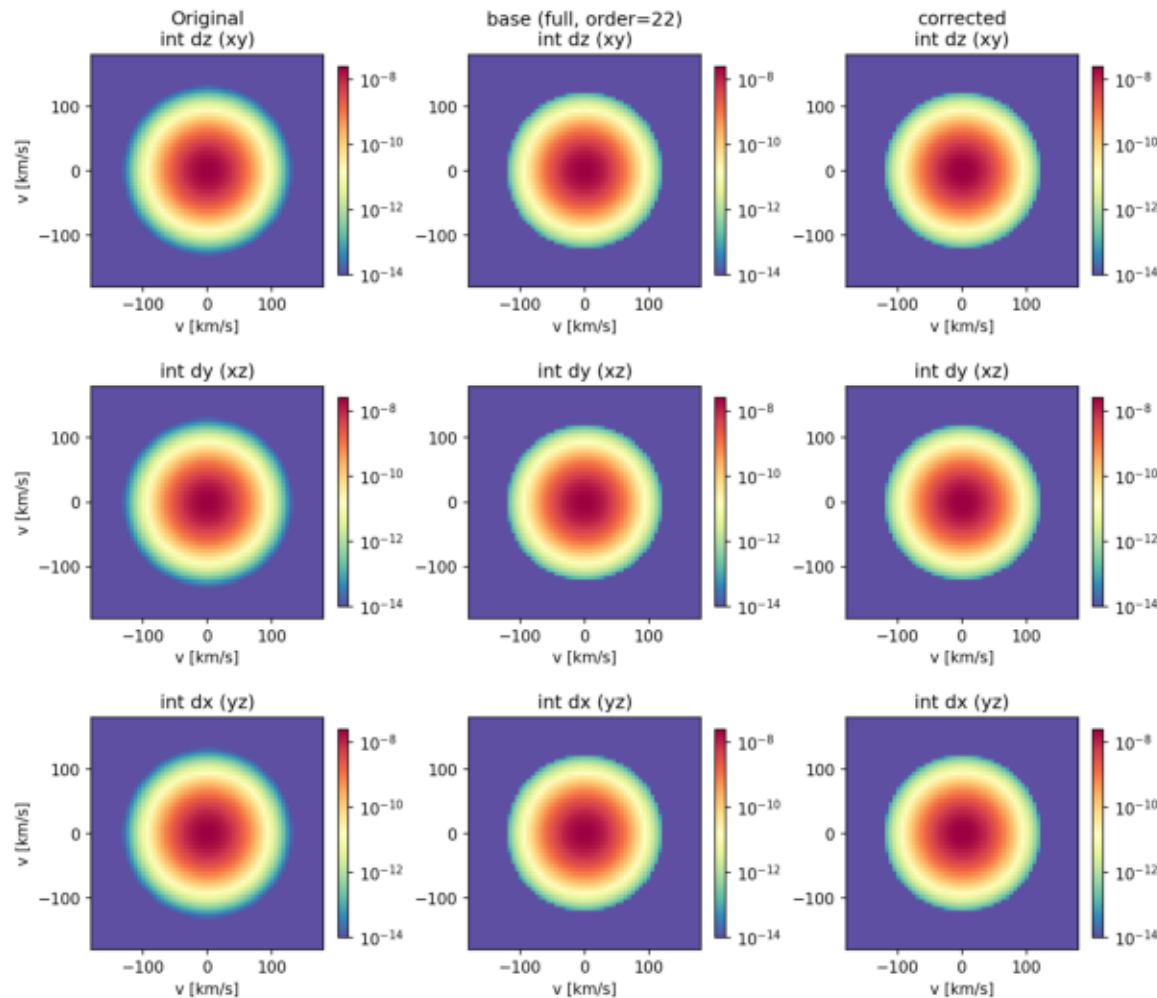
- Before (1 training snapshot): -28.0% mean improvement
- After (3 training snapshots): $+42.8\%$ mean improvement
- Best-checkpoint selection now uses the held-out timestep's
 - loss directly (true cross-timestep early stopping)



7. Cross-Timestep Fix — More Examples

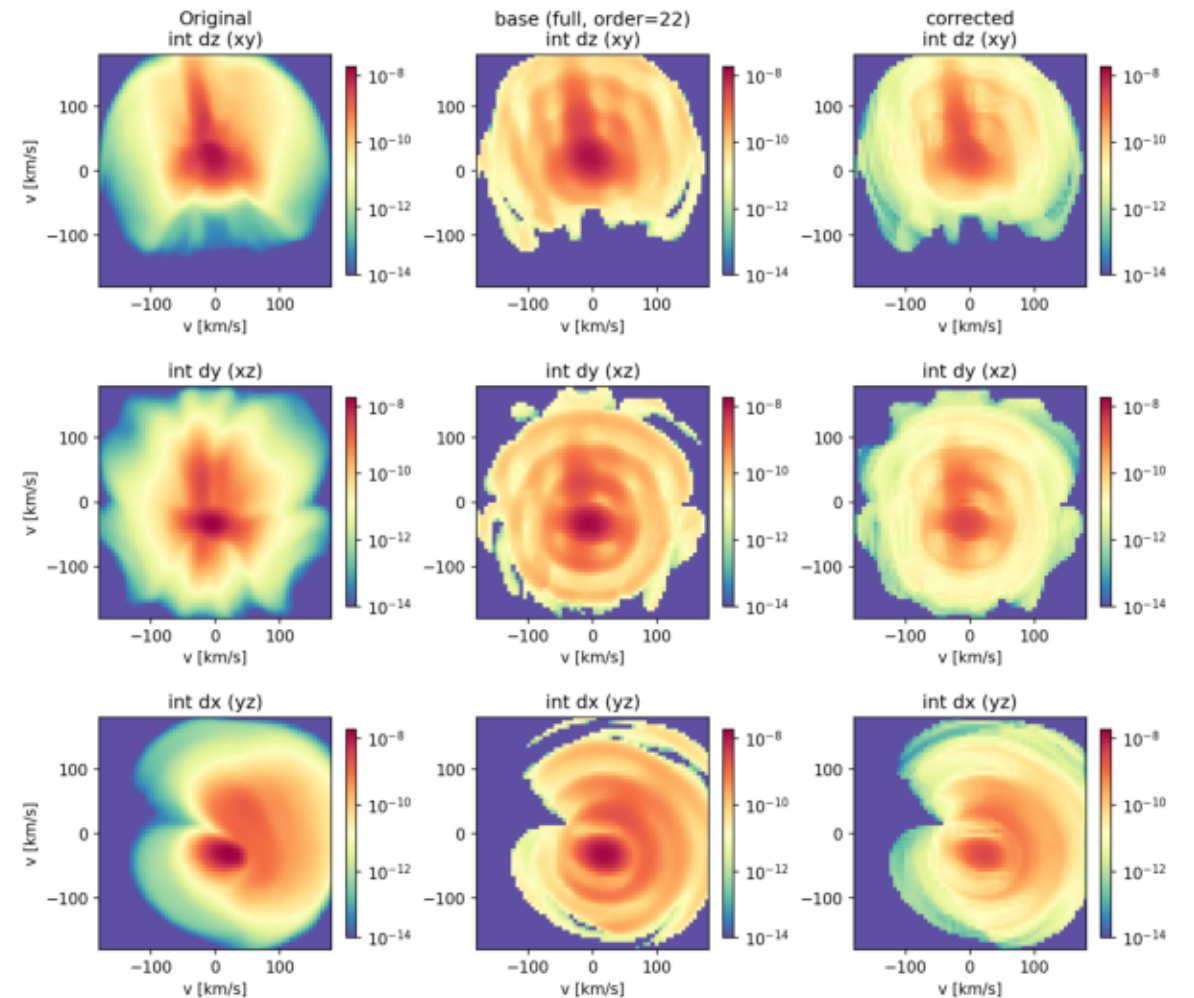
Unseen-timestep validation cells spanning the difficulty range

VDF before/after correction cid=16 (lowest_base (lobe-like)) eps_base=0.000 eps_corr=0.021



cid=16 — lobe-like, easiest

VDF before/after correction cid=1152 (highest_base (hardest)) eps_base=3.302 eps_corr=1.652



cid=1152 — hardest cell in the validation set

8. Conclusions and Open Weaknesses

- A small shared MLP, conditioned on cheap already-computed Hermite features, removes most Gibbs ringing without breaking positivity
- Combining local spatial context (patch) with global shape context (1-D spectra) was necessary — neither alone solved both failure modes
- Multi-snapshot training is REQUIRED for real generalization: a single training snapshot looks great in-sample (62%) but fails out-of-sample (−28%)
- Weakest point in the architecture: the regularization strength (λ_{max} , λ_{power} , ref_var) is still tuned empirically, per dataset
- Next steps: broader multi-snapshot / multi-simulation training sets, automated regularization selection, from-scratch (non-PyTorch) inference reimplementations for in-situ use inside Vlasiator